

SELECTION REPORT

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SUMMARY

[Stage Details](#)

1. SELECTION DATA

Database	Metals plus
Table	MaterialUniverse
Subset	Metals
Reference	

2. SELECTION CRITERIA (SUMMARY)

Stage	Attribute	Constraints
1		
2	Yield strength (elastic limit) (ksi)	≥ 50
	AND Fatigue strength at 10 ⁷ cycles (ksi)	≥ 30
	AND Metal hot forming	Acceptable, Excellent
	AND Weldability	Good, Excellent
	AND Water (salt)	Acceptable, Excellent
	AND Pitting resistance equivalent number (PREN)	≥ 30
	AND Stress corrosion cracking	Susceptible, Slightly susceptible, Not susceptible
3	Mass per unit of strength	
	Cost per unit of strength	
4	Pitting resistance equivalent number (PREN)	
	Cost per unit of strength	

3. SELECTION RESULTS

Records passing: All Stages 25 of 1923
Ranked by: Pitting resistance equivalent number (PREN)
Ranked order: Low to high

Rank	Material	Pitting resistance equivalent number (PREN)
1	 Stainless steel, ferritic, AISI 446, annealed	23 - 31
2	 Stainless steel, austenitic, AISI 316LVM, cold worked	24.4 - 31.2
3	 Nickel-Fe-Cr alloy, HAYNES HR-120, annealed	25.4 - 44.2

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2025

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Rank	Material	Pitting resistance equivalent number (PREN)
4	 Stainless steel, super-austenitic, alloy 20, annealed	25.6 - 30.9
5	 Stainless steel, duplex, AISI 329, annealed	26.3 - 34.6
6	 Stainless steel, duplex, UNS S32003 (AL 2003), annealed	26.7 - 32.3
7	 Nickel-Fe-Cr alloy, INCOLOY 825, annealed	27.8 - 35.1
8	 Stainless steel, austenitic, AISI 216, annealed	28.1 - 39.9
9	 Stainless steel, austenitic, Nitronic 50, XM-19, cold drawn, wire (nitrogen strengthened)	28.7 - 39.8
10	 Stainless steel, austenitic, Nitronic 50, XM-19, annealed (nitrogen strengthened)	28.7 - 39.8
11	 Stainless steel, duplex, AISI 2205, annealed	30.5 - 37.8
12	 Nickel-Fe-Cr alloy, NIMONIC 942, bar	32.3
13	 Nickel-Fe-Cr alloy, PYROMET 860, bar	32.4
14	 Nickel-Fe-Cr alloy, D-979, bar	32.6 - 38.3
15	 Nickel-Fe-Cr alloy, NIMONIC PE11, bar	32.7 - 38
16	 Iron-base-superalloy, N-155, solution treated	33.2 - 42.2
17	 Stainless steel, duplex, UNS S32550, annealed	35.2 - 43.9
18	 Stainless steel, duplex, UNS S44660, annealed	35.5 - 41.4
19	 Stainless steel, duplex, UNS S44635, annealed	35.9 - 41.8
20	 Stainless steel, ferritic, UNS S44700, annealed	38.8 - 45.2
21	 Stainless steel, duplex, UNS S44800, annealed	38.8 - 45.2
22	 Stainless steel, duplex, UNS S32750, annealed	39.5 - 46.1
23	 Stainless steel, ferritic, AL 29-4C, annealed	39.9 - 44.6
24	 Stainless steel, austenitic, AL-6XN, annealed	42.7 - 49.1
25	 Stainless steel, austenitic, UNS S31254, annealed	44.1 - 51.4

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PROJECT SELECTION SETTINGS:

Pass records with no data: Off

Fail estimated records: Off

[Change number of records to display...](#)

[Stage Details](#)

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STAGE 1:	TREE	Summary
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Linked Records	Pass
 Metals and alloys	1799

Records passing: 1799 of 1923

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STAGE 2:	SCREENING	Summary
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CONSTRAINTS

Attribute	Constraints
Yield strength (elastic limit) (ksi)	≥ 50
AND Fatigue strength at 10^7 cycles (ksi)	≥ 30
AND Metal hot forming	Acceptable, Excellent
AND Weldability	Good, Excellent
AND Water (salt)	Acceptable, Excellent
AND Pitting resistance equivalent number (PREN)	≥ 30
AND Stress corrosion cracking	Susceptible, Slightly susceptible, Not susceptible

STAGE SELECTION SETTINGS:

Pass when: Any part of record within selection

Records passing: 25 of 1923

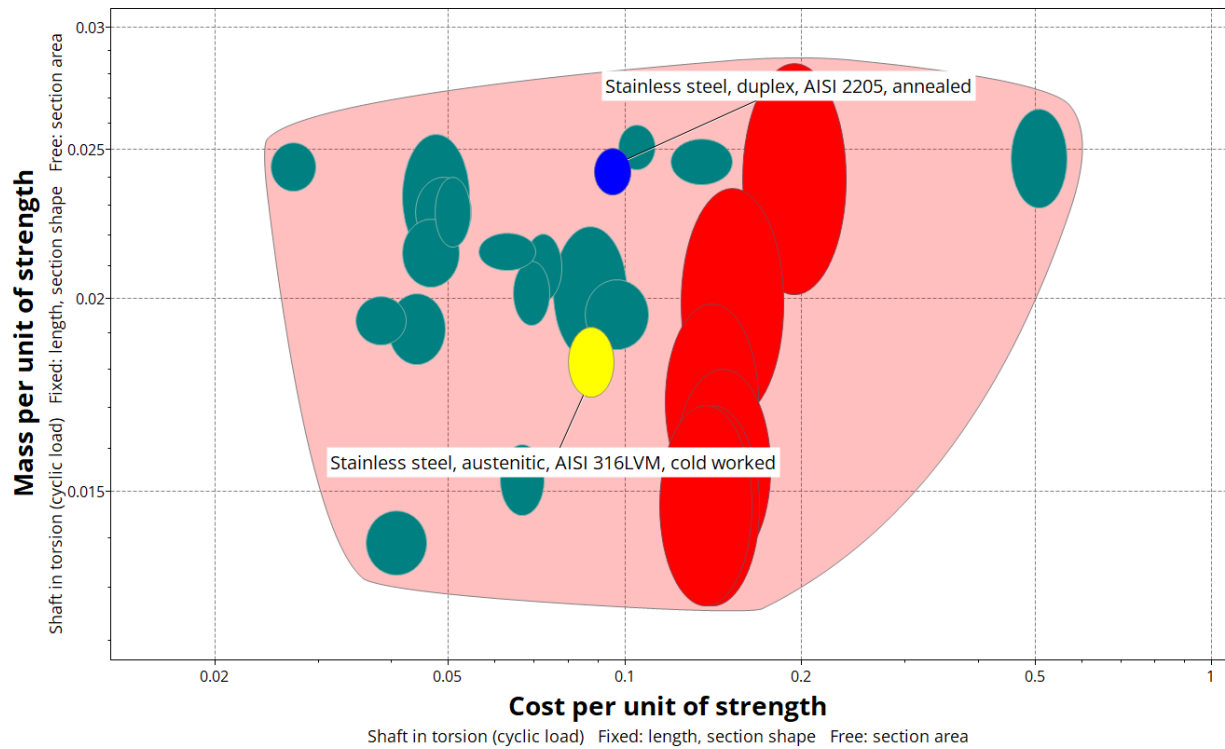
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STAGE 3:

COST PER UNIT OF STRENGTH, MASS PER UNIT OF STRENGTH

[Summary](#)



Axis attributes or formula

Mass per unit of strength

$[\text{Density}] / ([\text{Fatigue strength at } 10^7 \text{ cycles}]^{2/3})$

Cost per unit of strength

$[\text{Price}] * [\text{Density}] / ([\text{Fatigue strength at } 10^7 \text{ cycles}]^{2/3})$

Stage selection settings:

Show results from all enabled stages: On

Pass when: Any part of record within selection

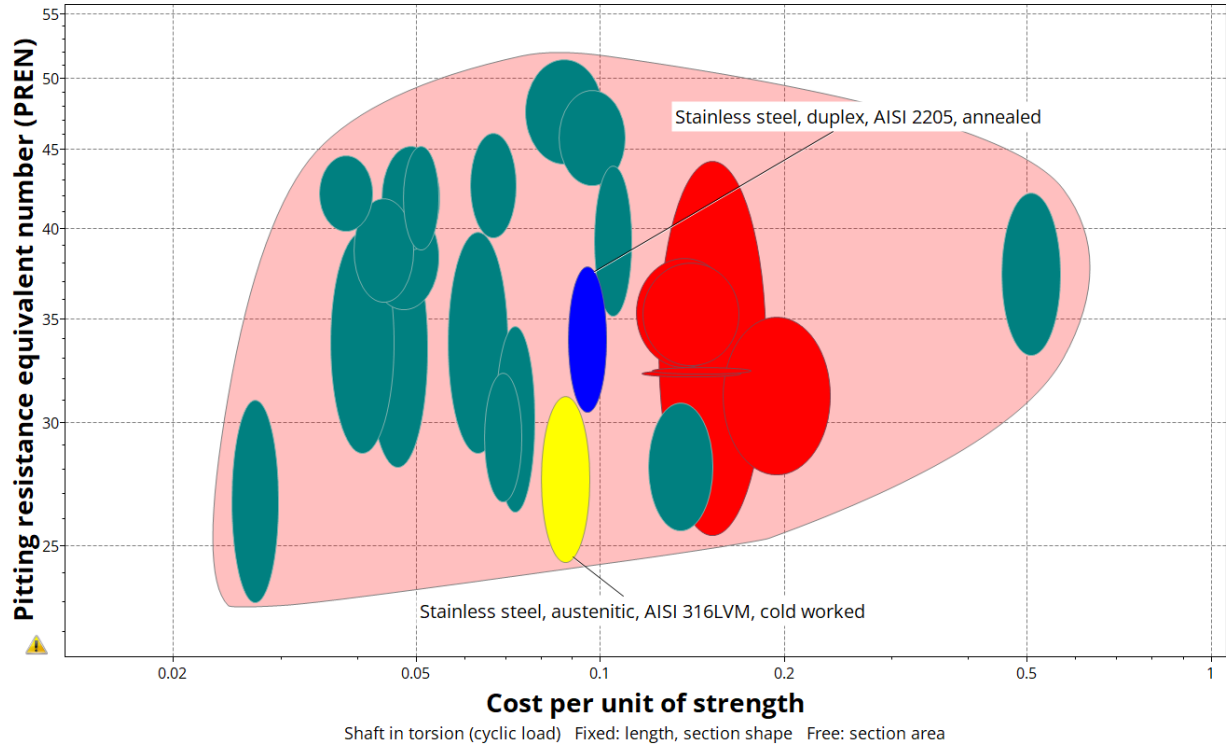
Records passing:

1912 of 1923

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STAGE 4:	COST PER UNIT OF STRENGTH, PITTING RESISTANCE EQUIVALENT NUMBER (PREN)	Summary
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Axis attributes or formula

Cost per unit of strength

$[Price] * [Density] / ([Fatigue\ strength\ at\ 10^7\ cycles]^{2/3})$

Stage selection settings:

Show results from all enabled stages: On

Pass when: Any part of record within selection

Records passing:

235 of 1923

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4. COMPARISON TABLE

	Stainless steel, austenitic, AISI 316LVM, cold worked	Stainless steel, duplex, AISI 2205, annealed	Nickel-Cr-Mo alloy, ILLIUM 98, as cast
Computed Properties			
Mass per unit of strength	0.0173 - 0.0192	0.0234 - 0.025	0.0314 - 0.0426
Cost per unit of strength	0.0802 - 0.0956	0.0888 - 0.102	0.429 - 0.598
General information			
Condition	50% cold worked	Solution annealed	As cast
UNS number	S31673	S31803 - S32205	
US name	S31673 to ASTM F139, ASTM F138	ASTM S32205, ASTM S31803, ASTM F60, ASTM F51	
EN name		X2CrNiMoN22-5-3	
EN number	1.4441 - X2CrNiMo18- 15-3, BS 7252	1.4462	
ISO name		X2CrNiMoN22-5-3	
GB (Chinese) name	00Cr18Ni14Mo3, 00Cr18Ni15Mo3N to GB/T 4234	0Cr21Ni5Ti to GB 3281, 00Cr18Ni5Mo3Si2 to GB/T 14976, GB/T 14975, GB 4237, GB 1220	
JIS (Japanese) name	SUS 316L	SUS329J3L	
Included in Materials Data for Simulation		Yes	
Materials Data for Simulation name		Stainless steel, 2205, annealed	
Composition overview			
Material family	Metal (ferrous)	Metal (ferrous)	Metal (non-ferrous)
Base material	Fe (Iron)	Fe (Iron)	Ni (Nickel)
Composition detail (metals, ceramics and glasses)			
C (carbon) (%)	0 - 0.03	0 - 0.03	0 - 0.07
Cr (chromium) (%)	17 - 19	21 - 23	26 - 30
Cu (copper) (%)	0 - 0.5	0	4 - 6.5
Fe (iron) (%)	59.4 - 67.8	63.7 - 71.9	0 - 1.5
Mn (manganese) (%)	0 - 2	0 - 2	0 - 1.5
Mo (molybdenum) (%)	2.25 - 3.2	2.5 - 3.5	7.5 - 9

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N (nitrogen) (%)	0 - 0.1	0.08 - 0.2	0
Ni (nickel) (%)	13 - 15	4.5 - 6.5	50.9 - 62.5
P (phosphorus) (%)	0 - 0.025	0 - 0.03	0
S (sulfur) (%)	0 - 0.01	0 - 0.02	0
Si (silicon) (%)	0 - 0.75	0 - 1	0 - 0.5
Price			
Price (USD/lb)	4.47 - 5.17	3.7 - 4.18	12.9 - 14.9
Price per unit volume (USD/ft ³)	2220 - 2590	1800 - 2040	6680 - 7820
Physical properties			
Density (lb/in ³)	0.287 - 0.29	0.282 - 0.283	0.3 - 0.303
Mechanical properties			
Young's modulus (10 ⁶ psi)	26.9 - 27.5	28.3 - 29.7	23.9 - 28.3
Specific stiffness (lb _f .ft/lb)	7.76e6 - 7.93e6	8.36e6 - 8.77e6	6.62e6 - 7.83e6
Yield strength (elastic limit) (ksi)	107 - 141	66.7 - 74	37 - 46.4
Tensile strength (ksi)	130 - 172	92.8 - 138	47.9 - 59.5
Specific strength (lb _f .ft/lb)	30900 - 40800	19700 - 21800	10200 - 12800
Elongation (% strain)	12 - 21.3	20 - 25	10 - 30
Tangent modulus (ksi)	290	306	98.8
Compressive modulus (10 ⁶ psi)	27 - 27.4		
Compressive strength (ksi)	107 - 141	66.7 - 74	37 - 46.4
Flexural modulus (10 ⁶ psi)	26.9 - 27.5	28.3 - 29.7	23.9 - 28.3
Flexural strength (modulus of rupture) (ksi)	105 - 127	66.7 - 74	37 - 46.4
Shear modulus (10 ⁶ psi)	9.71 - 10.7	10.9 - 12.3	8.7 - 11.6
Shear strength (ksi)	97.8 - 130		
Bulk modulus (10 ⁶ psi)	26.4 - 26.8		16.7 - 22.5
Poisson's ratio	0.33	0.295 - 0.305	0.27 - 0.29
Shape factor	26	47	28
Hardness - Vickers (HV)	306 - 357	248 - 310	150 - 190
Hardness - Rockwell B (HRB)		99 - 108	
Hardness - Rockwell C (HRC)	32 - 37	22 - 31	
Hardness - Brinell (HB)	278 - 324	278 - 308	

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Elastic stored energy (springs) (ft.lbf/in ³)	17.8 - 30.1	6.39 - 7.87	2.19 - 3.46
Fatigue strength at 10 ⁷ cycles (ksi)	58.5 - 68.2	37.9 - 41.9	18.9 - 29.7
Fatigue strength model (stress amplitude) (ksi) #	48 - 83.1	27.6 - 57.4	17.3 - 32.3
Impact & fracture properties			
Fracture toughness (ksi.in ^{0.5})	76.4 - 92.8	131 - 177	95.2 - 106
Toughness (G) (ft.lbf/in ²)	18 - 26.2	50.4 - 89	28.3 - 36.7
Ductility index (mil)	0.00494 - 0.00544	0.0179 - 0.0219	0.0269 - 0.0329
Thermal properties			
Melting point (°F)	2730	2550 - 2640	2250 - 2520
Maximum service temperature (°F)	572 - 594	635 - 689	1500 - 2010
Minimum service temperature (°F)	-328	-76 - -58	-459
Thermal conductivity (BTU/ft.hr.°F)	8.09 - 9.42	8.32 - 9.01	4.62 - 6.93
Specific heat capacity (BTU/lb. °F)	0.116	0.115 - 0.119	0.0955 - 0.107
Thermal expansion coefficient (μstrain/°F)	9.17	7.5 - 10.1	5.56 - 8.33
Thermal shock resistance (°F)	460 - 599	270 - 358	216 - 329
Thermal distortion resistance (BTU/hr.ft)	882000 - 1.03e6	853000 - 1.16e6	631000 - 1.09e6
Latent heat of fusion (BTU/lb)	112 - 123	163 - 197	120 - 129
Electrical properties			
Electrical resistivity (μohm.in)	29.1	30.9 - 32.1	47.2 - 55.1
Electrical conductivity (%IACS)	2.33	2.11 - 2.2	1.23 - 1.44
Galvanic potential (V)	-0.15 - -0.07	-0.26 - -0.18	-0.12 - -0.04
Magnetic properties			
Magnetic type	Non-magnetic	Magnetic	Non-magnetic
Optical, aesthetic and acoustic properties			
Transparency	Opaque	Opaque	Opaque
Softness to touch	6.21 - 6.47	6.33 - 6.71	7.3 - 7.87
Warmth to touch	12.6 - 13.6	12.9 - 13.5	15.2 - 18.9
Acoustic velocity (in/s)	190000 - 192000	197000 - 202000	175000 - 191000

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Mechanical loss coefficient (tan delta)	0.00095 - 0.0013	0.0006 - 0.00074	0.0006 - 0.001
Healthcare & food			
Food contact	Yes	Yes	Yes
Medical grades? (USP Class VI, ISO 10993)	Yes		
Healthcare applications	Bone fixation and repair, Catheters and cannulas, Implantable pacemakers and defibrillators, Joint replacement, Spinal devices, Stents, Surgical instruments, Wound and tissue closure		
Sterilizability (ethylene oxide)	Excellent		
Sterilizability (radiation)	Excellent		
Sterilizability (steam autoclave)	Excellent		
Guidance for MRI Safety	High Risk for Potential Interaction	Caution - Strong Interaction	Low Risk for Potential Interaction
Restricted substances risk indicators			
RoHS 2 (EU) compliant grades?	Yes	Yes	Yes
EU REACH Candidate List indicator (0-1, 1 = high risk)	0	0	0
SIN List indicator (0-1, 1 = high risk)	0	0	0
Critical materials risk			
Abundance risk level	Medium	Medium	Medium
Sourcing and geopolitical risk level	Low	Low	Low
Environmental country risk level	Low	Low	Low
Price volatility risk level	Medium	Medium	Low
Conflict material risk level	None	None	Caution
Processing properties			
Metal casting	Unsuitable	Unsuitable	Acceptable
Metal cold forming	Excellent	Limited use	Unsuitable
Metal hot forming	Excellent	Acceptable	Unsuitable
Metal press forming	Excellent	Limited use	Unsuitable

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Metal deep drawing	Excellent	Limited use	Unsuitable
Machining speed (sfm)	36	52	25
Weldability	Excellent	Good	Good
Weldability_Notes	Preheating is not required, post weld heat treatment is required	Preheating and post weld heat treatments are not required	Preheating and post weld heat treatments may be required
Weldability - MIG	Excellent	Good	
Weldability - plasma	Excellent	Good	
Weldability - SAW	Excellent	Good	
Weldability - TIG	Excellent	Good	
Brazeability	Good	Fair	
Carbon equivalency	1.22 - 1.59	1.29 - 1.66	
Durability			
Water (fresh)	Excellent	Excellent	Excellent
Water (salt)	Excellent	Excellent	Excellent
Weak acids	Excellent	Excellent	Excellent
Strong acids	Excellent	Acceptable	Excellent
Weak alkalis	Excellent	Excellent	Excellent
Strong alkalis	Excellent	Excellent	Excellent
Organic solvents	Excellent	Excellent	Excellent
Oxidation at 500C	Excellent	Excellent	Excellent
UV radiation (sunlight)	Excellent	Excellent	Excellent
Galling resistance (adhesive wear)	Acceptable	Limited use	Acceptable
Flammability	Non-flammable	Non-flammable	Non-flammable
Corrosion resistance of metals			
Pitting resistance equivalent number (PREN)	24.4 - 31.2	30.5 - 37.8	
Pitting and crevice corrosion resistance	Medium (20-30)	High (30-40)	
Stress corrosion cracking	Slightly susceptible	Not susceptible	Not susceptible
Stress corrosion cracking_Notes	Rated in chloride; Other susceptible environments: Hydrogen sulfide	Rated in chloride; Other susceptible environments: Hydrogen sulfide	Rated in chloride; Other susceptible environments: Hydroxide
Intergranular (weld line) corrosion resistance	Good	Excellent	

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Inorganic acids	Moderate	Good	
Organic acids	Good	Good	
Alkalies	Moderate	Moderate	
Humidity / water	Excellent	Excellent	
Sea water	Good	Good	
Sour oil and gas	Moderate	Good	
Environmental indicators - material production			
Climate change (CO ₂ -eq) (virgin grade production) (lb/lb)	7.59 - 8.37	6.22 - 6.85	15.6 - 17.2
Climate change (CO ₂ -eq) (typical grade production) (lb/lb)	5.65 - 6.23	4.77 - 5.25	10 - 11.1
Embodied energy (virgin grade production) (BTU/lb)	38400 - 42400	29500 - 32500	113000 - 125000
Embodied energy (typical grade production) (BTU/lb)	29200 - 32200	23000 - 25400	70000 - 77100
Water usage (in ³ /lb)	4210 - 4650	3990 - 4430	8280 - 9130
Environmental indicators - processing			
Metal casting, climate change (CO ₂ -eq) (lb/lb)			3.22 - 3.81
Metal casting, embodied energy (BTU/lb)			19400 - 22700
Metal rolling and forging, climate change (CO ₂ -eq) (lb/lb)	1.24 - 1.78	0.812 - 0.985	
Metal rolling and forging, embodied energy (BTU/lb)	8180 - 11700	5340 - 6460	
Metal extrusion, hot, climate change (CO ₂ -eq) (lb/lb)	2.41 - 3.47	1.56 - 1.89	
Metal extrusion, hot, embodied energy (BTU/lb)	15900 - 22800	10200 - 12400	
Wire drawing, climate change (CO ₂ -eq) (lb/lb)	0.287 - 0.317	0.287 - 0.317	
Wire drawing, embodied energy (BTU/lb)	1880 - 2080	1880 - 2080	
Metal powder forming, climate change (CO ₂ -eq) (lb/lb)	5.92 - 6.54	5.48 - 6.52	
Metal powder forming, embodied energy (BTU/lb)	31200 - 34500	28900 - 34400	

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Chemical vapor deposition (CVD), climate change (CO2-eq) (lb/lb)	793 - 960	1150 - 1540	854 - 1010
Chemical vapor deposition (CVD), embodied energy (BTU/lb)	4.14e6 - 5.02e6	6.02e6 - 8.05e6	4.46e6 - 5.28e6
Machining, coarse, climate change (CO2-eq) (lb/lb)	1.31 - 1.57	1.18 - 1.33	1.07 - 1.22
Machining, coarse, embodied energy (BTU/lb)	8610 - 10300	7760 - 8730	7050 - 7980
Machining, fine, climate change (CO2-eq) (lb/lb)	2.02 - 2.59	1.63 - 1.88	1.31 - 1.54
Machining, fine, embodied energy (BTU/lb)	13300 - 17000	10700 - 12300	8590 - 10100
Machining, grinding, climate change (CO2-eq) (lb/lb)	2.73 - 3.62	2.08 - 2.43	1.54 - 1.86
Machining, grinding, embodied energy (BTU/lb)	17900 - 23700	13700 - 16000	10100 - 12200
Machining, laser, CO2, climate change (CO2-eq) (lb/ft)	0.0634 - 0.0768	0.0922 - 0.123	0.0683 - 0.0808
Machining, laser, CO2, embodied energy (BTU/in)	27.6 - 33.4	40.1 - 53.7	29.7 - 35.2
Recycling and end of life			
Recycle	Yes	Yes	Yes
Functional recycle	Yes	No	No
Climate change (CO2-eq), recycling (lb/lb)	2.05 - 2.26	1.68 - 1.85	2.03 - 2.24
Embodied energy, recycling (BTU/lb)	11900 - 13100	9140 - 10100	7910 - 8730
Recycle fraction in current supply (%)	35	32	41
Combust for energy recovery	No	No	No
Landfill	Yes	Yes	No
Biodegrade	No	No	No
Geo-economic data for principal component			
Principal component	Iron	Iron	Nickel
Typical exploited ore grade (%)		45.1 - 49.9	0.997 - 1.1
Minimum economic ore grade (%)		25 - 70	0.1 - 2

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Abundance in Earth's crust (ppm)	41000 - 63000	41000 - 63000	59 - 90
Abundance in seawater (ppm)	0.0025 - 0.003	0.0025 - 0.003	0.0005 - 0.002
Annual world production, principal component (ton/yr)	2.19e9	2.19e9	2.33e6
Reserves, principal component (l. ton)	1.56e11	1.56e11	8.38e7
Links			
Elements in this material	0	5	5
Global Metals similar alloys	110	163	3
JAHM Curve data	113	16	0
ProcessUniverse	127	137	112
Producers	34	30	18
Reference	38	44	48
Shape	24	24	24
Full Datasheet			

# FUNCTIONAL PARAMETERS	
Stress Ratio	-1
Number of Cycles (cycles)	1e7